Cascading Failures in Power Grids: A Replication Study of Dynamic vs. Static Models

Niklas Bacher

Abstract

This project replicates the main results of (Schäfer et al., 2018), which analyze cascading failures in power grids using a dynamical model based on the swing equation. In contrast to static approaches, the model captures transient dynamics that arise following disturbances and may trigger additional failures.

Using the Spanish high-voltage transmission network as well as additional networks from France and Great Britain, the replication compares cascade sizes under static and dynamical assumptions. The results confirm that cascade outcomes depend strongly on the location of the initial failure and that dynamical models produce substantially larger cascades than static models across a wide range of parameter values.

1 Introduction

Large-scale power outages highlight the vulnerability of modern electricity systems to cascading failures. A recent example is the 2025 blackout on the Iberian Peninsula, where an initial disturbance triggered a sequence of failures that propagated across large parts of the Spanish and Portuguese transmission grid, leading to widespread supply disruptions affecting millions of consumers. While the precise causes are still under investigation, early reports suggest that the event involved a combination of line overloads and dynamic instabilities that amplified the initial shock. Such events illustrate how local failures can trigger system-wide consequences, raising the question of how reliably these cascades can be predicted.

Existing approaches to cascading failures in power grids often rely on static models that approximate power flows using steady-state assumptions (Crucitti et al., 2004; Pahwa et al., 2014). In these models, failures occur when line capacities are exceeded under a given load distribution. While this approach is computationally convenient, it neglects the transient dynamics that arise immediately after a disturbance. In reality, power systems are governed by synchronization dynamics. Sudden perturbations can induce short-term oscillations in frequency and power flows before the system reaches a new equilibrium (Filatrella et al., 2008), hence triggering additional failures that would not be predicted by static models (Schäfer et al., 2018).

The paper by (Schäfer et al., 2018) addresses this limitation by explicitly modeling the dynamic response of the grid using a network of coupled oscillators. In this project, the main results of the authors are replicated using the authors' modeling framework.

2 Results/Main Analysis

To reproduce the main findings of (Schäfer et al., 2018), I implement the authors' dynamical model of cascading failures based on the swing equation, using Python instead of Mathematica. In this framework, each node in the power grid is modeled as a coupled oscillator characterized by its phase angle and frequency:

$$\frac{d\theta_i}{dt} = \omega_i \tag{1}$$

$$\frac{d\omega_i}{dt} = P_i - \gamma\omega_i + \sum_j K_{ij} \sin(\theta_j - \theta_i) \tag{2}$$

where θ_i denotes the phase angle of node i, ω_i its angular velocity, P_i the net power injection and K_{ij} the coupling strength between nodes. This formulation captures the synchronization dynamics of the grid and allows the system to respond dynamically to perturbations (Filatrella et al., 2008; Kundur et al., 1994).

Based on these dynamics, the power flow along a energy transmission line (i, j) in the grid at time t is defined as:

$$F_{ij}(t) = K_{ij} \sin(\theta_j(t) - \theta_i(t)) \tag{3}$$

A line fails when the absolute power flow exceeds its capacity, which is defined as a fraction of the coupling strength:

$$|F_{ij}(t)| > C_{ij} = \alpha K_{ij} \tag{4}$$

Here, K_{ij} represents the coupling strength between two nodes, which can be interpreted as the maximum amount of power that can be transmitted along the line under ideal conditions. The parameter $\alpha \in [0, 1]$ defines the tolerance level of the network and determines how much of this maximum capacity can be safely used. The resulting capacity $C_{i,j}$ therefore reflects a practical limit below the theoretical maximum, accounting for safety constraints such as overheating or stability requirements (Schäfer et al., 2018).

If the flow on a line exceeds this threshold at any point in time, the line is considered overloaded and is removed from the network. This removal redistributes the power flows across the remaining lines, which may in turn become overloaded. By iteratively removing overloaded lines and updating the system dynamics, cascading failures can emerge as a sequence of dependent outages.

2.1 Cascading failures in the Spanish power grid

As in the original paper, the Spanish high-voltage transmission network provided in the authors' supplementary materials is used. The network consists of nodes representing generators and consumers, connected by transmission lines with specified coupling

strengths K_{ij}. Power injections P_i are assigned following the original setup, ensuring that the system is initially in a stable operating state.

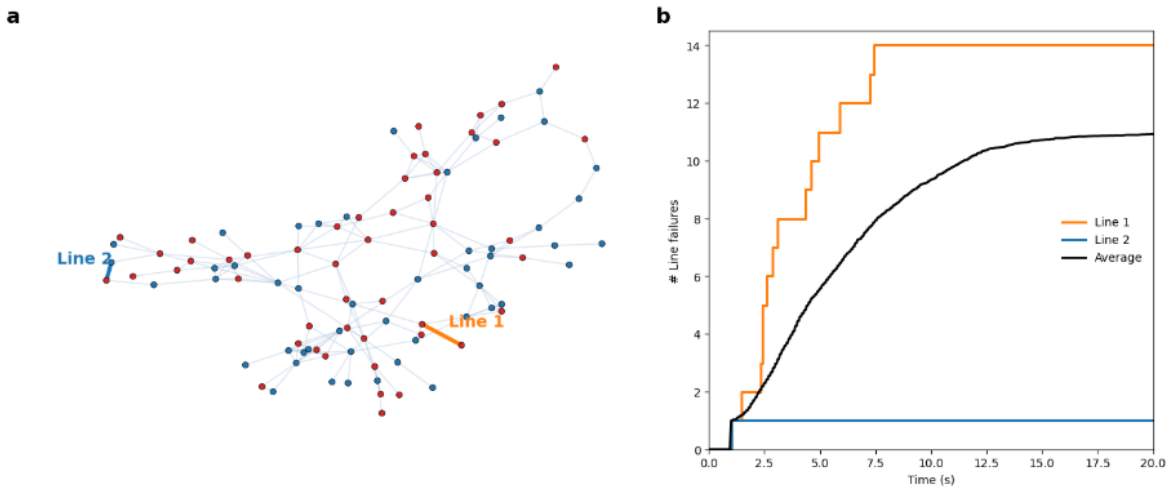

Figure 1: Cascading failures in the Spanish power grid.

Figure 1 illustrates the propagation of cascading failures after the removal of a single transmission line. Panel (a) shows the network structure, highlighting the initially failed line and the subsequent spread of failures across the grid. Panel (b) displays the temporal evolution of the number of failed lines for two representative initial outages, as well as the average cascade size.

2.2 Static vs. dynamic cascading failures

To further assess the role of transient dynamics, I compare cascade sizes obtained from the dynamical model with those from a static cascade model. Figure 2 shows the fraction of failed lines as a function of the tolerance parameter α for both approaches.

Panel (a) shows that cascade sizes decrease as the tolerance parameter α increases, indicating that higher line capacities make the network more robust to failures. However, across almost the entire range of α, the dynamical model produces substantially larger cascades than the static model. This difference is particularly pronounced at intermediate and low values of α, where static cascades remain relatively small while dynamical cascades still affect a large fraction of the network.

Panel (b) provides additional insight by showing the fraction of unsynchronized nodes. For low values of α, the system loses synchronization almost completely, indicating a breakdown of stable operation. As α increases, synchronization is gradually restored.

2.3 Cascade size distributions across network configurations

Robustness of cascading failure patterns is further examined by replicating the distribution of cascade sizes for different network configurations introduced by (Schäfer et al., 2018). Results are reported in Figure 4 in the Appendix.

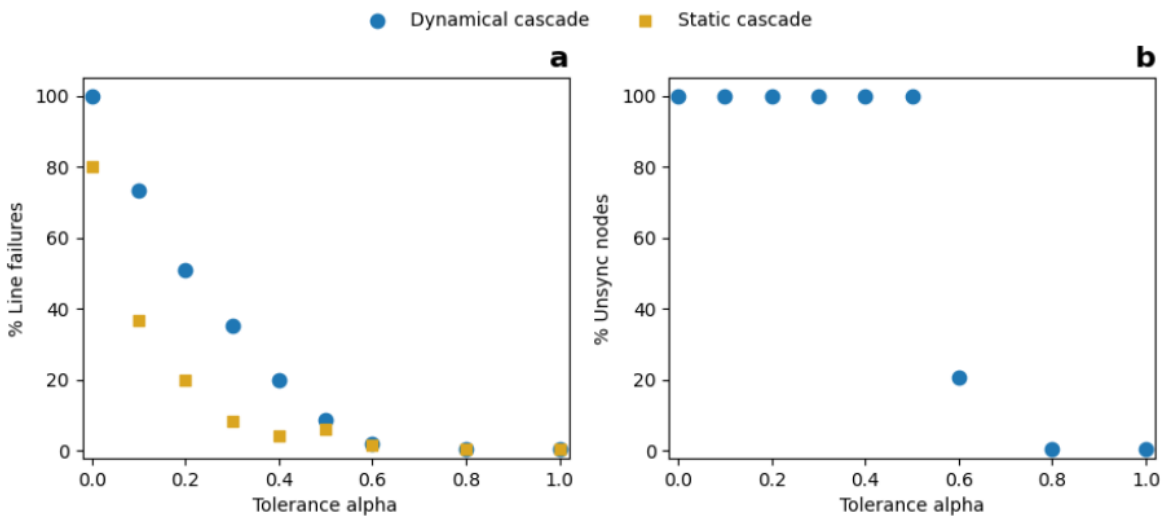

Figure 2: Comparison of cascading failures in static and dynamical models as a function of the tolerance parameter α.

These configurations differ in how power generation and transmission capacities are distributed across the network. In the *distributed* case, power generation is spread relatively evenly across nodes, leading to a more homogeneous structure. In contrast, a *centralized* configuration concentrates generation in a small number of nodes, increasing the load on specific transmission paths. A *heterogeneous* setting introduces variation in coupling strengths K_{ij}, resulting in uneven transmission capacities across the network.

In addition to the Spanish network, the analysis is extended to transmission networks from France and Great Britain provided in the original dataset, testing whether the observed cascade patterns generalize beyond a single national grid. The same modeling framework and parameter settings are applied to all networks to ensure comparability. Results for France and Great Britain are reported in the Appendix.

3 Discussion

The replication results confirm the central finding of (Schäfer et al., 2018): transient dynamics play a crucial role in determining cascading failures in power grids. In particular, the comparison between static and dynamical models shows that static approaches tend to underestimate cascade sizes, as they do not capture temporary overloads that arise during the adjustment process following a disturbance.

At the same time, the results highlight that cascade outcomes depend strongly on the location of the initial failure, as shown in Figure 1. Some line removals lead to only minor disruptions, while others trigger large cascades affecting a substantial fraction of the network. This heterogeneity suggests that vulnerability is not uniformly distributed but instead concentrated in specific parts of the network.

From a methodological perspective, the replication closely follows the original mod-

eling framework, but some practical limitations arise. In particular, the computation of cascade size distributions for different network configurations proved to be computationally expensive. To address this, the analysis in Section 2.3 is based on a reduced sample of initial line failures. While this approach allows for reasonable computation, it may introduce some sampling variability in the estimated distributions. However, the overall qualitative patterns remain consistent with the original findings.

These results are relevant for the assessment of power grid stability, as they indicate that models relying solely on steady-state assumptions may underestimate systemic risk. In the real-world, transient dynamics occur on short time scales and can trigger failures that are not captured by static analyses. Incorporating such dynamics may therefore improve the identification of critical components and the evaluation of network resilience. (Schäfer et al., 2018)further develop such a mechanism to identify critical lines, which is not replicated here due to scope constraints.

At the same time, the analysis relies on a simplified representation of power grids. The swing equation captures the synchronization dynamics of generators and the resulting power flows, but it simplifies from several features of real-world systems. In particular, the model does not explicitly account for detailed protection mechanisms such as relay settings or automatic load shedding, which can interrupt cascading processes or, in some cases, accelerate them (Kundur et al., 1994). In real systems, such mechanisms may disconnect overloaded components early or stabilize the network through controlled interventions, thereby altering the progression of cascades.

Overall, the replication supports the main conclusions of (Schäfer et al., 2018) and demonstrates that the role of transient dynamics in cascading failures can be reproduced using an independent implementation.

References

Crucitti, P., Latora, V., and Marchiori, M. (2004). Model for cascading failures in complex networks. *Physical Review E*, 69(4):045104.

Filatrella, G., Nielsen, A. H., and Pedersen, N. F. (2008). Analysis of a power grid using a kuramoto-like model. *The European Physical Journal B*, 61(4):485–491.

Kundur, P., Balu, N. J., and Lauby, M. G. (1994). *Power System Stability and Control*. McGraw-Hill.

Pahwa, S., Scoglio, C., and Scala, A. (2014). Abruptness of cascade failures in power grids. *Scientific Reports*, 4:3694.

Schäfer, B., Witthaut, D., Timme, M., and Latora, V. (2018). Dynamically induced cascading failures in power grids. *Nature Communications*, 9(1):1975.

Appendix

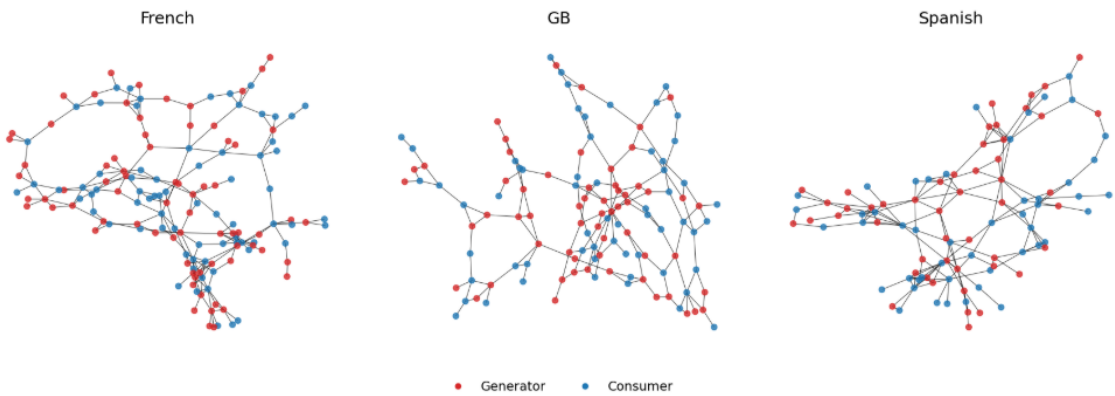

Figure 3: Network layouts of the national power-grid topologies of France, Great Britain and Spain. The positions are obtained from a force-directed layout and therefore do not correspond to geographical coordinates, but are used to visualize the structural organization and connectivity patterns of the three grids.

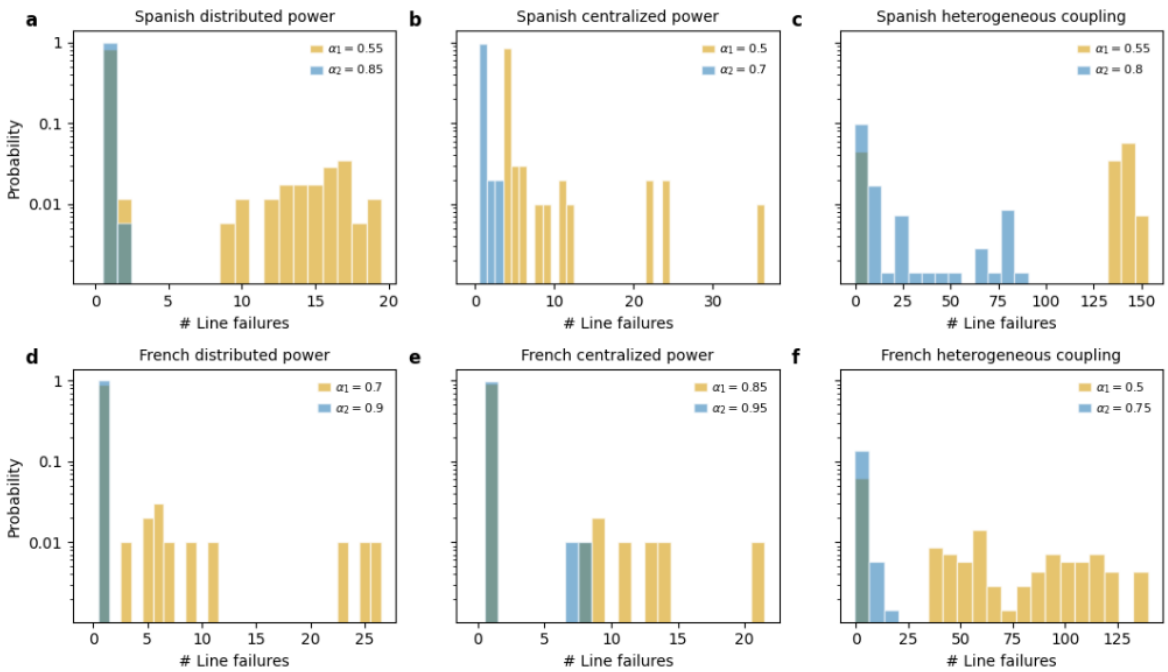

Figure 4: Distributions of cascade sizes, measured as the number of failed transmission lines, for the Spanish and French power-grid models under distributed power, centralized power and heterogeneous coupling. In each panel, the histograms compare two tolerance values α_1 and α_2, where larger α corresponds to a higher overload threshold and typically leads to smaller cascades. No α could be obtained for the British grid, as no information was available in the author's paper or source code.